# Perioperative Thromboprophylaxis: A Clinical Pharmacy Perspective

## Preventing DVT and PE in Surgical Patients

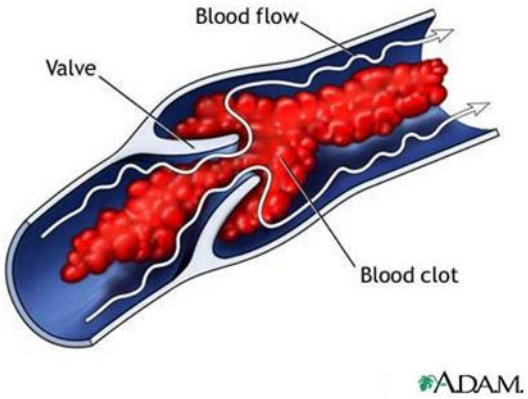
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# Introduction

- **Key Points**:
  - Deep venous thrombosis (DVT) is a common and preventable perioperative complication.
  - Risk varies by surgery type:
    - **Neurosurgery**: 24%
    - **Total Hip Replacement**: 45-57%
    - **Total Knee Replacement**: 40-84%
    - **Hip Fracture Surgery**: 36-60%.



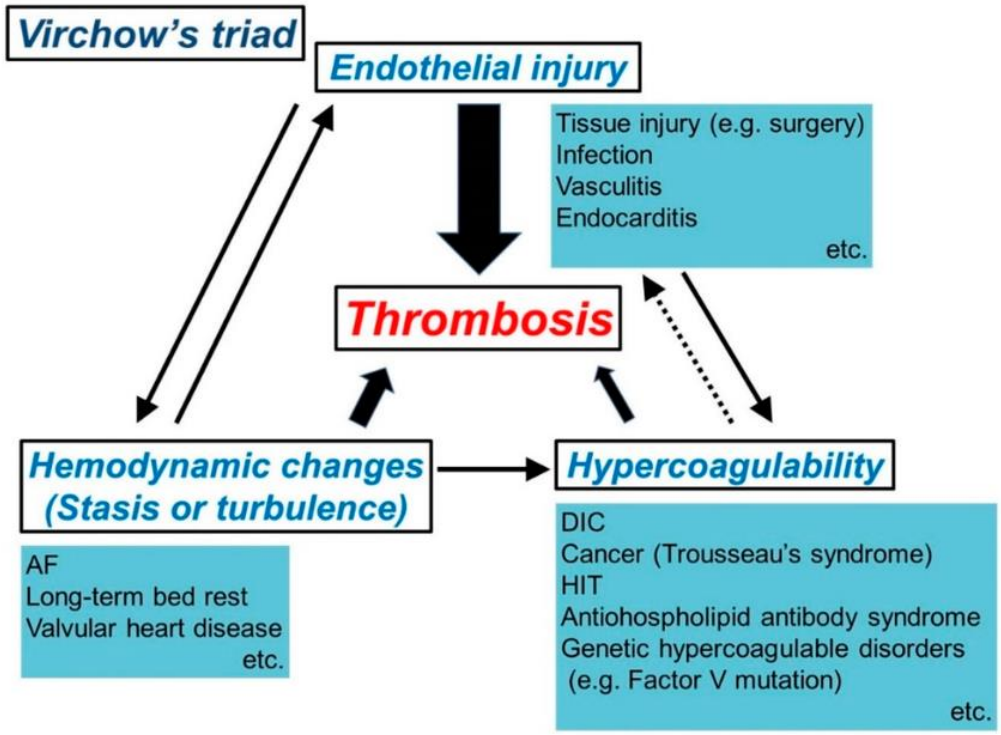

  - **Goal**: Prevent DVT and pulmonary embolism (PE).

# Pathophysiology

Virchow triad

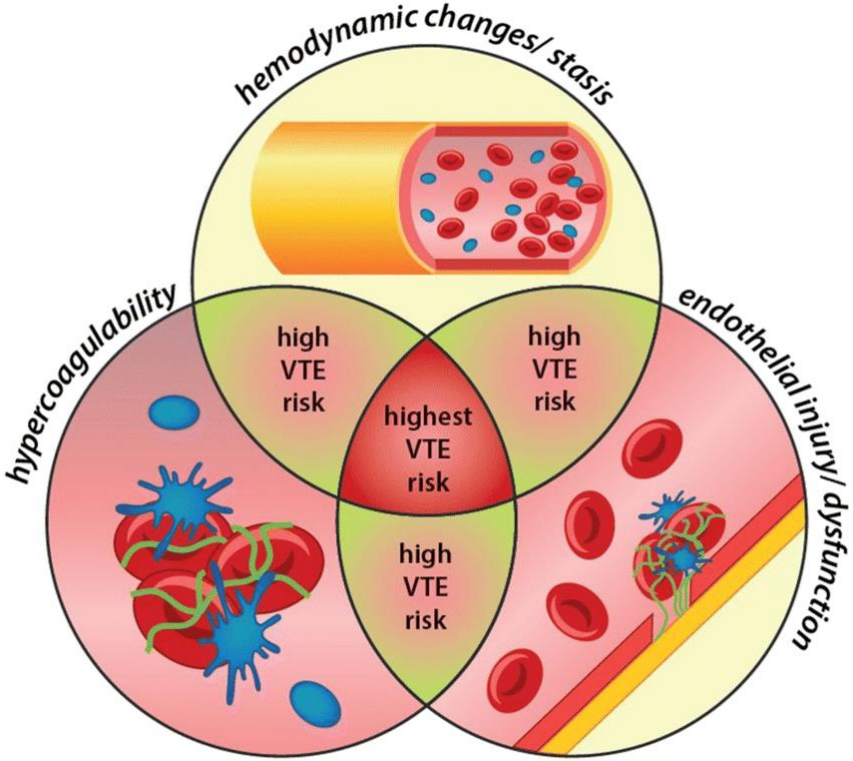
# Pathophysiology of DVT

- **Virchow's Triad**:
  - **Stasis**: Venous pooling due to immobility and anesthesia.
  - **Hypercoagulability**: Increased procoagulant factors.
  - **Intimal Injury**: Trauma to blood vessels (dt ↑VD with anesthesia, surgery)
- **Consequences**: DVT → PE → Morbidity and mortality.



# Risk Factors for DVT

- **Patient-Related**:
  - Age >50 years, obesity, history of DVT/PE, cancer, diabetes, heart failure, Immobility, varicose veins, hypercoagulable states
- **Surgery-Related**:
  - Major surgery, prolonged anesthesia, orthopedic surgery.



# Classification

The Seventh American College of Clinical Pharmacy (ACCP) Consensus Conference recommend that patients are classified as having low, moderate, high, and very high risks for the development of DVT or PE

# Risk Stratification (ACCP Guidelines)

| Risk Level | Patient Characteristics | DVT Risk | PE Risk |
|---|---|---|---|
| **Low** | • Minor surgery, age <40, no risk factors, GA < 30 min | 2% | 0.20% |
| **Moderate** | • Major surgery, age <40, no risk factors, GA > 30 min<br>• Any surgery in age 40-60 with no risk factors<br>• minor surgery with risk factors | 10-20% | 1-2% |
| **High** | • Major surgery, age >60, no risk factors<br>• Any surgery in age 40-60 with risk factors<br>• MI | 20-40% | 2-4% |
| **Highest** | • prior venous thromboembolism<br>• Malignant disease<br>• Hypercoagulable state<br>• elective major lower extremity orthopedic surgery<br>• Hip fracture<br>• Stroke<br>• Polytrauma | 40-80% | 4-10% |

# Classify the surgery

**High risk**

- eg:
  - LL arthroplasty
  - osteotomy

**Low risk**

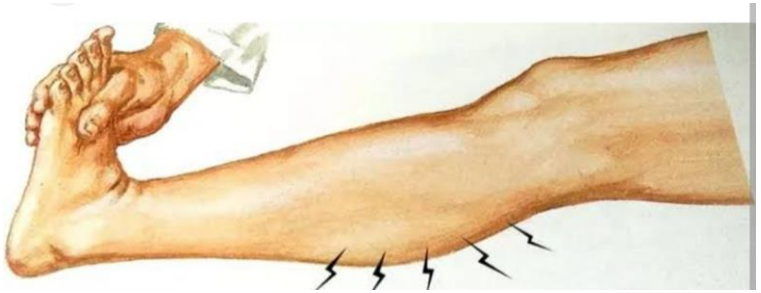
- eg:
  - Knee scope
  - Foot
  - UL

# Clinical picture

# Clinical Presentation

- **DVT Symptoms**:
  - Leg pain, swelling, redness, warmth, tenderness.
  - **Homan's Sign**: Calf pain on dorsiflexion.
- **PE Symptoms**:
  - **Small PE**: Asymptomatic.
  - **Medium PE**: Dyspnea, hemoptysis, chest pain.
  - **Large PE**: Syncope, cardiac arrest.



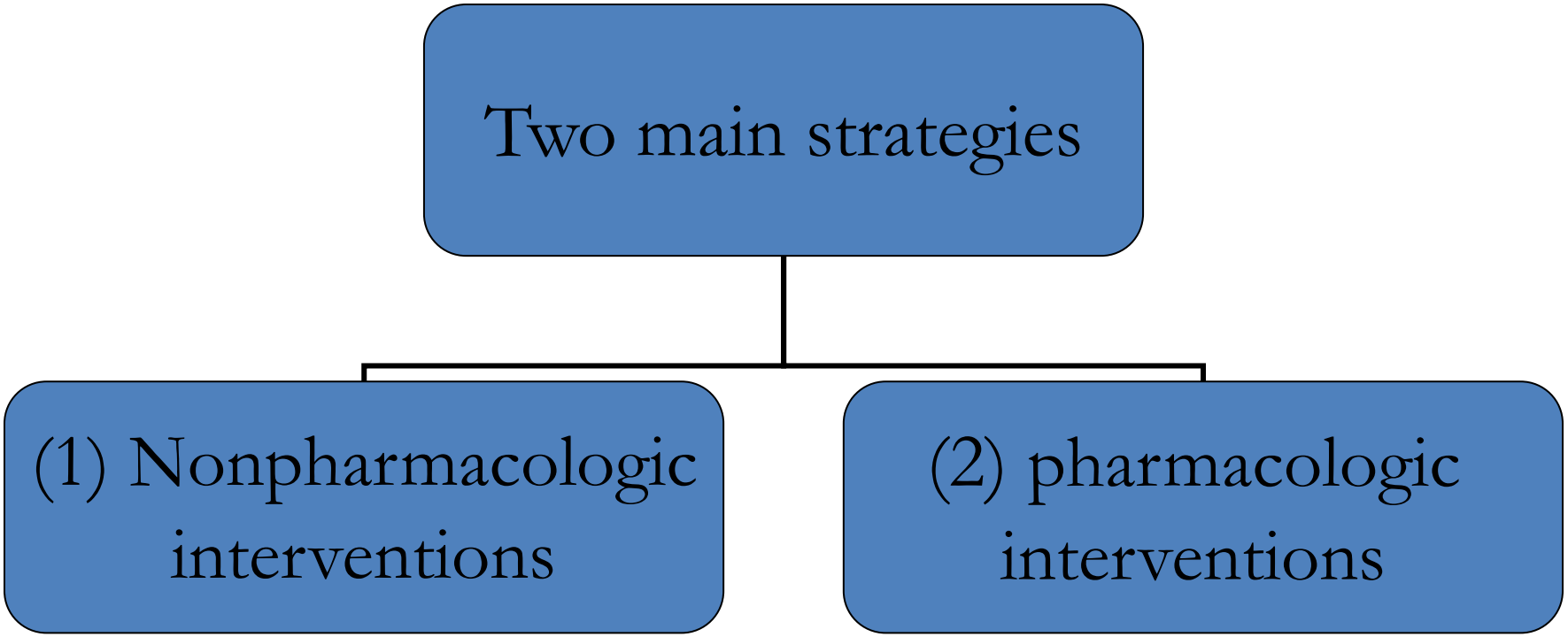
# Investigations

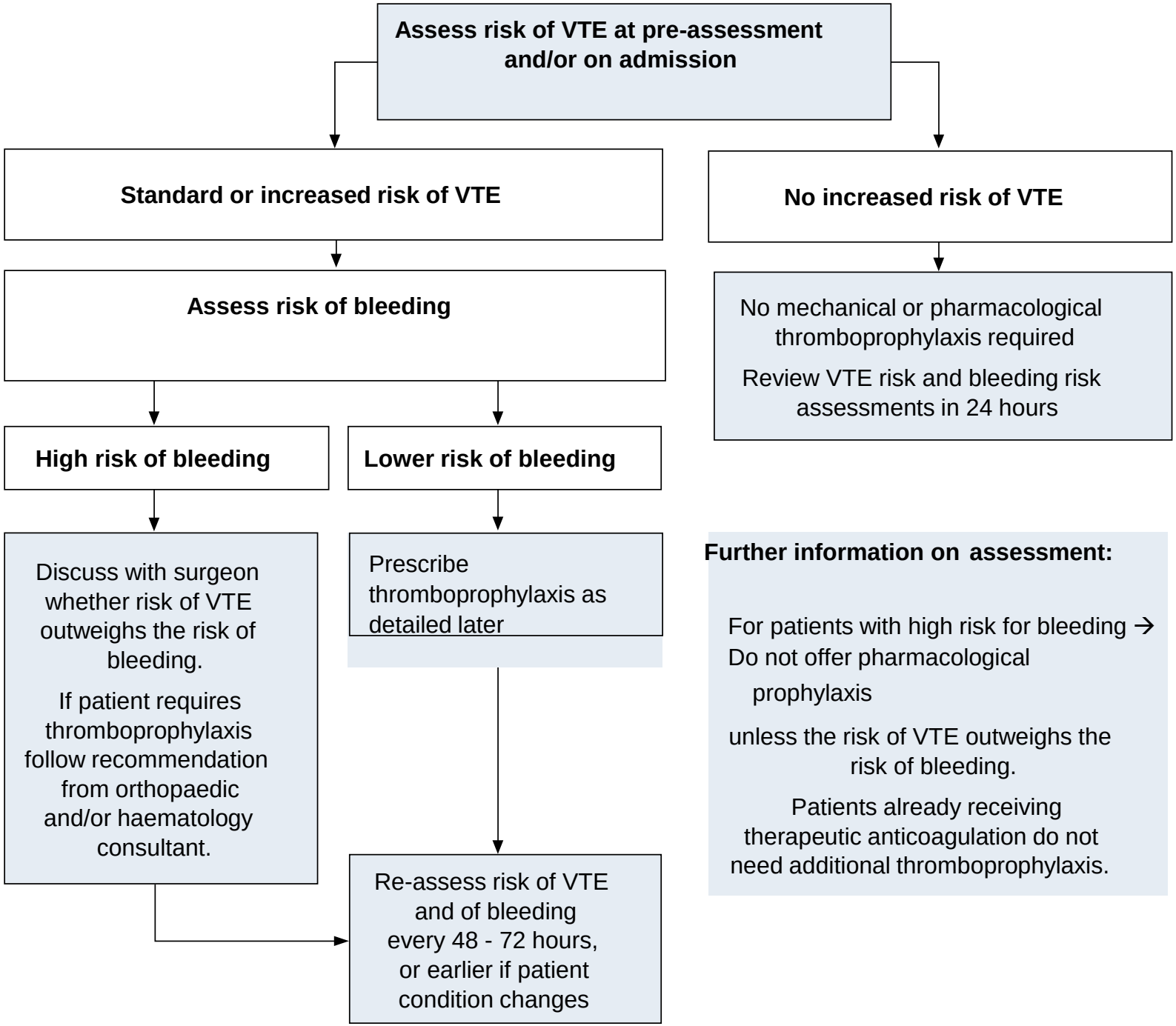
- **Diagnostic Tools**:
  - **Doppler Ultrasound**: For DVT.
  - **CT Pulmonary Angiography**: For PE.
  - **D-Dimer**: Elevated in thrombosis (non-specific).

# Management

# Goals of therapy:

1. Prevention of the occurrence of DVT or PE.
2. Prevention of the consequences of DVT or PE.




Assess risk of VTE at pre-assessment and/or on admission
Standard or increased risk of VTE
No increased risk of VTE
Assess risk of bleeding
No mechanical or pharmacological thromboprophylaxis required
Review VTE risk and bleeding risk assessments in 24 hours
High risk of bleeding
Lower risk of bleeding
Discuss with surgeon whether risk of VTE outweighs the risk of bleeding.
If patient requires thromboprophylaxis follow recommendation from orthopaedic and/or haematology consultant.
Prescribe thromboprophylaxis as detailed later
Re-assess risk of VTE and of bleeding every 48 - 72 hours, or earlier if patient condition changes
Further information on assessment:
For patients with high risk for bleeding → Do not offer pharmacological prophylaxis
unless the risk of VTE outweighs the risk of bleeding.
Patients already receiving therapeutic anticoagulation do not need additional thromboprophylaxis.

# Risks of bleeding

- surgery expected within the next 12 hours
- active bleeding or risk of bleeding including
    - new-onset stroke
    - platelet count < 75 x 109/L
    - acute liver failure
    - active duodenal or gastric ulcer
- concurrent use of therapeutic anticoagulant
- acute bacterial endocarditis
- any spinal intervention

  (prophylactic enoxaprin dose is contraindicated for 12 hours before spinal and epidural anaesthetics and lumbar puncture. Enoxaparin contraindicated for 4 hours after spinal and epidural anaesthetics and removal of epidural catheter)
- persistant uncontrolled hypertension (BP ≥ 230/120 mmHg)
- untreated inherited bleeding disorder (e.g. hemophilia or von Willebrand's)
- spinal surgery (seek specialist advice)
- proliferative diabetic retinopathy

# Recommendation for thromboprophylaxis

- **Every patient must be risk assessed and recorded**

- **Patients on long term antiplatelets prior to surgery:**

  – on one single long term antiplatelet → prescribing prophylactic LMWH for up to five weeks post orthopedic surgery is acceptable in most cases.

  –on a combination of antiplatelet agents → consult hematology or cardiology.

- **Patients on oral anticoagulants prior to surgery:**

  –anticoagulants can be restarted as soon as clinically appropriate. Patients on warfarin may require LMWH until therapeutic INR has been achieved

## ELECTIVE PROCEDURES

| Procedure | Risk stratification | During admission | On discharge |
|---|---|---|---|
| **Lower limb arthroplasty or other major lower limb elective surgery (inc ankle surgery and hindfoot surgery)** | Standard VTE risk | Enoxaparin SC 40mg once daily (*) (**) | Continue enoxaparin SC for an overall treatment course of 5 weeks or<br>Aspirin orally 150mg daily for overall treatment course of 5 weeks<br>If patient requires a cast/ boot/ splint post-op, continue thromboprophylaxis for the duration of cast/ boot/ splint |
| | Increased VTE risk | | Continue enoxaparin SC for an overall treatment course of 5 weeks<br>If patient requires a cast/ boot/ splint post-op, continue thromboprophylaxis for the duration of cast/ boot/ splint |
| **Other elective surgery (inc upper limb, arthroscopy and forefoot surgery)** | Standard VTE risk | No need for pharmacological thromboprophylaxis | |
| | Increased VTE risk | Follow recommendations from orthopaedic and/ or haematology consultant | |
| **Elective spinal surgery** | Thromboprophylaxis assessment done on a case-by-case basis depending on the type of surgery and risk factors – follow recommendations from spinal surgeon | | |

| TRAUMA PROCEDURES | | | |
|---|---|---|---|
| **Procedure** | **Risk stratification** | **During admission** | **On discharge** |
| **Hip fracture surgery** | Standard VTE risk | Enoxaparin SC 40mg once daily (*) (**) | Continue enoxaparin SC for an overall treatment course of 2 weeks or until discharge (whichever is sooner) |
| | Increased VTE risk | | Continue enoxaparin SC for an overall treatment course of 5 weeks |
| **Lower limb fracture surgery (inc ankle fracture surgery and hindfoot surgery)** | Standard VTE risk | Enoxaparin SC 40mg once daily (*) (**) | Continue enoxaparin SC for an overall treatment course of 5 weeks or<br>Aspirin orally 150mg daily for 5 weeks<br>If patient requires a cast/ boot/ splint post-op, continue thromboprophylaxis for the duration of cast/ boot/ splint |
| | Increased VTE risk | | Continue enoxaparin SC for an overall treatment course of 5 weeks<br>If patient requires a cast/ boot/ splint post-op, continue thromboprophylaxis for the duration of cast/ boot/ splint |
| **Pelvic fracture surgery** | Enoxaparin SC 40mg once daily for a total of 5 weeks (*) (**) | | |
| **Spinal fracture surgery** | Thromboprophylaxis assessment done on a case-by-case basis depending on the type of surgery and risk factors – follow recommendations from spinal surgeon | | |
| **Spinal cord injury** | Enoxaparin SC 40mg once daily (**) | | |
| **Multiple trauma** | Thromboprophylaxis assessment done on a case-by-case basis depending on the extent of injuries – follow recommendations from orthopaedic and/ or haematology consultant | | |

<table>
<tr><th colspan="4">NO SURGICAL PROCEDURE</th></tr>
<tr><td>Orthopaedic patients who do not require surgery nor a lower limb cast, boot or splint</td><td colspan="3">Enoxaparin SC 40mg once daily (**) - usually stopped at discharge or earlier if patient is no longer at high thrombotic risk when re-assessed</td></tr>
<tr><td rowspan="2">Lower limb casts, boots or splints</td><td>Standard VTE risk</td><td rowspan="2">Enoxaparin SC 40mg once daily (*) (**)</td><td rowspan="2">Risk assess every patient prior to discharge.<br>Manage it case by case</td></tr>
<tr><td>Increased VTE risk</td></tr>
</table>

(*) Started on the day of surgery at 6pm or at least 4 hours after surgery, which ever is latest

(**) Reduce dose to 20mg if eGFR <30ml/min or if patient weighs <50Kg.

- **Timing of enoxaparin administration**

  - *For in-patients with a significant reduction in mobility* ENOXAPARIN should be prescribed at 6pm the night before surgery. Otherwise it should be started after surgery at the later of: 4hrs post-operatively or 6pm. Then at 6pm on subsequent days.

  - *For patients admitted on the day of surgery* that require ENOXAPARIN thromboprophylaxis:
    - Anti-embolism stockings should be considered from admission
    - ENOXAPARIN started after surgery at the later of: 4hrs post-operatively or 6pm. Then at 6pm on subsequent days.

- **Precautions with Epidural & Spinal Anaesthetic Techniques**

  - Epidural and spinal anaesthetic techniques should not be carried out within 12hr of a prophylactic dose of ENOXAPARIN.

  - Epidural catheters should not be removed within 12 hr of a prophylactic dose of ENOXAPARIN.

  - Wait >4h after any of these procedures before giving next dose of ENOXAPARIN.

  - In most cases administration of ENOXAPARIN at 6pm will avoid any difficulties here.

- **Monitoring platelet count**

  - All patients prescribed heparin, including LMWH, should have a baseline platelet count assessed.

  - Postoperative patients receiving LMWH have a low risk of HIT and do not require routine platelet count monitoring for this purpose.

# Antiplatelet agents

- Antiplatelet agents are generally considered ineffective in preventing PE.
- Some guidelines advise against the use of aspirin alone in DVT prophylaxis.

# Nonpharmacologic measures

- recommended for low-risk patients throughout the perioperative period until they are ambulatory.
- Include: Graduated compression stockings (GCS), Intermittent pneumatic compression (IPC).
- Inferior vena cava filters are no longer recommended in primary prevention for perioperative DVT and PE.
- ***Early ambulation remains the most important nonpharmacologic approach to prevention of perioperative DVT and PE.***










THANK YOU